The Action Lags the Answer: An Agent's Tool Commitment Becomes Causally Steerable Deeper Than Its Verbalizable Answer, in Two Architectures

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Abstract

Anthropic Interpretability Team [2026] show that language models carry an emergent, verbalizable *global workspace* ("J-space"): a few dozen concepts, read out by a Jacobian lens, whose coordinate interventions causally reroute multi-hop *answers* at mid-network depth. Their study contains no agentic content, leaving open whether an agent's *action* selection is reachable at the same depth as its verbalizable answer. We test this on **two** open-weights reasoning agents—a Qwen3.6-27B (dense) coding agent and gpt-oss-20b (mixture-of-experts)—using a row-restricted J-lens estimator whose readout directions we validate with a specificity control (a steer flips a held answer 17/20 at half the residual norm, while random and unrelated directions flip 0/20 and an equal-magnitude random push does not even move the answer). With the instrument validated, we find a **robust depth lag**: the agent's tool commitment becomes causally steerable via the verbalizable direction *strictly deeper* than the verbalizable answer. On Qwen the answer is steerable from the mid-workspace (L27, 42% depth; 11/20 vs random 0) where the action is not (0/60, *worse* than random 21), the action first steerable at the workspace tail (L47, 73%, 60/60). On gpt-oss the answer's onset is later (the tail, 14/20 vs random 0) and the action's later still (only the motor band, 40/40 vs random 4; not the tail, 0/40). The absolute onset depths are model-dependent; the lag is consistent across the two architectures tested—on both there is a band where the verbalizable answer is steerable (vs. a magnitude-matched random-direction control) and the action is not (the gpt-oss decision point is a weaker probe, edit-commit fidelity 0.14 vs 0.23, so it corroborates rather than leads). The action's readout also emerges late (AUROC peaks 0.74 at L47 in Qwen), and projecting out the top-10 verbalizable directions at the decision point leaves the commitment differential intact (0.174 → 0.167). We *localize* (not discover) the knowledge–action gap [Basu et al., 2026] as this answer→action depth lag; the contribution is the lag, its cross-architecture replication, and the null ablation. Scoped implication: a J-lens-style verbalizable monitor reads a reasoning agent's answer at a shallower depth—and in a band before—its action commitment. Ledger, vectors, pre-registration, and a recompute-every-number evaluation are released; the readout curve recomputes offline to 10^{-6} (16/16 layers) and the dissociation numbers reproduce under an independent re-implementation.

1 Introduction

A reasoning agent that decides to call an irreversible tool has, somewhere in its forward pass, committed to that action. Where? Anthropic Interpretability Team [2026] give a sharp new vocabulary for "where": a Jacobian lens (J-lens) reveals an emergent *global workspace*, a small set of verbalizable concepts occupying a middle band of depth, whose coordinates can be swapped to causally change a model's multi-hop *answer*. Ablating the top workspace directions destroys multi-hop reasoning while sparing surface parsing. The paper studies pretrained-style completions; it contains no agent, tool-use, or action content, leaving whether *action selection or tool routing runs through the same workspace, and at the same depth,* an open question for deployed agents.

Our prior arc supplies the exact complementary prior. For a long-horizon coding agent we showed the control of an action lives not at the mid-layer "task is done" verdict but in a late, task-matched commitment band ∼30 layers downstream [Vicentino, 2026a], that this late lever brakes irreversible actions [Vicentino, 2026b], and that a reasoning model's chain-of-thought is causal for its action only in that same late band [Vicentino, 2026c]. In the workspace vocabulary this predicts a specific, falsifiable claim: the agent's action commitment becomes reachable by the verbalizable direction *later in depth* than the answer. We test it directly.

Contribution. On *two* open-weights reasoning agents (a dense 27B and an MoE 20B) we establish a robust depth lag: the verbalizable-workspace direction that causally reroutes a held *answer* at some depth does *not* reroute the agent's *tool commitment* at that same depth (it is at or below a random control there); the action becomes specifically steerable only strictly deeper. The absolute onset depths differ by model (mid-workspace vs the tail for the answer; tail vs motor for the action), but on both there is a band where the answer is steerable and the action is not. Ablating the verbalizable subspace at the decision point does not disturb the commitment. We are careful about novelty (§2): we neither discover the knowledge–action gap nor claim action is merely deeper than the answer—both are known. The new object is the *dissociation plus null ablation*, and its scoped monitoring implication.

2 Related work and novelty

A post-result literature scan returns no work testing whether an agent's action commitment routes through the *emergent* workspace; three neighbors bound our claims and are differentiated here. Basu et al. [2026] document a knowledge–action gap—internal probes separate cases the model fails to act on (98% vs 45%)—but treat it as an actionability limit; we do not claim the gap, we *localize* it as an answer→action depth lag and show the mid-workspace direction that reroutes the answer is not yet reachable through the action there. Sun et al. [2026] read *whether* a tool is needed with a linear probe; we study *which* tool is committed and its causal routing, and show it is dissociated from the workspace direction and survives its ablation. Our nearest neighbor is Wu et al. [2026], who show which-tool choice is linearly readable early and only causally *steerable* at a late layer, in dense transformers (Gemma 3, Qwen 3, Qwen 2.5, Llama 3.1); we do not claim that early-readable/late-steerable ordering, which is theirs. Our contribution is orthogonal to it: the answer-vs-action *depth lag* via the shared emergent *verbalizable* (J-space) direction—not a bespoke tool direction—and its extension to an MoE architecture (gpt-oss-20b), not among the dense families they test. Sun & Kazakov [2026] decode tool-call dependency topology from the residual stream while explicitly disclaiming behavioral control, and Cui & Luo [2026] show semantic direction forms early while deep layers stabilize outputs in an MoE; because tool-related signals are linearly *readable* early [Wu et al., 2026, Sun & Kazakov, 2026], we scope our claim to the *specific causal steerability of the which-tool commitment via the verbalizable direction*, not readability in general. Global-workspace models of cognition in networks [Goyal et al., 2021, VanRullen & Kanai, 2021] *impose* a workspace as an architectural module; we test an *emergent* one and ask whether action routes through it.

3 Method

Model and decision points. Qwen3.6-27B (64 layers; depth% = $\ell/64$). From 99 SWE-bench Pro trajectories [Vicentino, 2026b] we take 60 deterministic `edit` (`str_replace_editor`) and 60 `bash` decision points, prefill-only, following the protocol of Vicentino [2026a]. Answers are a constructed 20-item few-shot two-hop QA set (single-token answers; 20/20 baseline accuracy).

Row-restricted J-lens estimator. For a single-token concept v and layer ℓ we estimate the readout row $g_{v,\ell} = \partial\,\mathrm{logit}_v(\text{final position})/\partial h_{\ell,\text{final}}$ by one backward pass per (v, prompt), parameters frozen and the embedding made the sole gradient leaf, averaged over an on-distribution estimation set. This is the diagonal (same-position) readout direction, not the full Jacobian matrix; we do not claim a complete reproduction of the J-lens.

Instrument validation (specificity control). The pre-registered coordinate *swap* $h \leftarrow h + V(\sigma(c)-c)$ is a near-no-op in this basis (the raw-gradient projections are near-equal, $c_{\text{correct}} \approx c_{\text{alt}}$), so we validate and use a *steer* along the normalized $(g_{\text{alt}} - g_{\text{correct}})$ direction scaled to $\alpha\|h\|$—a disclosed deviation to a second Anthropic-used intervention on the same directions. Table 1 shows the steer is specific: at α=0.5 (aggregated over the L27–55 workspace band) it flips a held answer to the designated alternative 17/20, while unrelated-pair and random directions flip 0/20 and a random push of equal magnitude does not move the answer at all. (The per-band answer steer in Table 2 restricts this to each band separately.) (A readout-correlation check is uninformative here: RMSNorm is degree-0 homogeneous, so by Euler's theorem $g_v \cdot h \approx 0$ by construction.)

α	contrast→alt	unrelated→alt	random→alt	random moved off correct
0.5	**17/20**	0/20	0/20	0/20
1.0	**20/20**	0/20	0/20	19/20

Table 1: Specificity control: the estimated readout direction is causal and specific; equal-magnitude random pushes do not reroute (and at α=0.5 do not even move) the answer.

4 Results

H1: the action's verbalizable readout emerges late (workspace tail). Figure 2 plots the AUROC of $(g_{\text{edit}} - g_{\text{bash}}) \cdot h$ separating edit- from bash-decision points by depth. It is near chance through most of mid-workspace, first exceeding chance only near L35, and rises to a peak of 0.74 at layer 47 (73%), sustaining 0.59–0.69 into the motor band. The action's verbalizable readout only becomes usably discriminative at the tail; it is near chance through most of the mid-workspace where the answer is already steerable. (The curve recomputes offline from the public vectors and residuals to 10^{-6} at all 16 layers.)

H2 and the dissociation: answers reroute in mid-workspace; the action does not. We steer the commitment along $(g_{\text{other}} - g_{\text{taken}})$ at the decision points (α=0.5), band-wise, against a random-direction control, and—parallel and apples-to-apples —steer the *answer* per band (Figure 3, Table 2). At mid-workspace the verbalizable direction *specifically* reroutes answers (11/20 vs random 0) but the action steer is *worse than random* (0/60 vs 21/60): no specific workspace-mediated rerouting of the commitment. Specific action control appears at the tail (60/60 vs random

23) and completes at motor—so the action is not absent from the workspace, it is reachable only from the workspace *tail* onward, strictly deeper than the answer.[1] Because the identical magnitude at the identical layers reroutes the answer but not the action, the mid-workspace action result is not a magnitude/disruption artifact. Decoding the mid-workspace action steer confirms this: the global argmax is dominated by structural tokens (> 47/60, newline 8/60; only 1/60 a tool token, 6 distinct tokens), i.e. the mid-workspace intervention perturbs emission without producing a clean committed tool call. All positive dissociations are significant against their magnitude-matched random controls (Fisher exact: answer mid-workspace $p{=}1.5 \times 10^{-4}$; action tail $p{=}3.8 \times 10^{-15}$, motor $p{=}2.8 \times 10^{-16}$).

band	answer: contrast→alt (random)	action: edit→bash (random)
mid-workspace (L27–43)	**11/20 (0)**	**0/60 (21)**
tail (L47–55)	20/20 (0)	60/60 (23)
motor (L59–63)	20/20 (0)	60/60 (21)

Table 2: The dissociation. Same steer, same magnitude, same layers: the verbalizable direction reroutes the answer in mid-workspace but not the action.

H3: the commitment survives ablation of the verbalizable subspace. Projecting out the top-10 verbalizable directions (by lens activation on the agent distribution) at the decision points leaves the task-appropriate commitment differential essentially unchanged: $P(\text{edit})$ at edit points $0.474 \to 0.473$, at bash points $0.299 \to 0.306$; differential $0.174 \to 0.167$ (from unrounded values; Figure 4). The commitment does not live in the verbalizable subspace at the decision position—the load-bearing rebuttal to "you are re-reading an early-decodable tool signal." This is an instance of the well-documented detect≠control / knowledge–action pattern [Basu et al., 2026] for the verbalizable direction specifically, not a fresh discovery that the commitment is non-verbalizable; the novelty rests on the depth-lag dissociation and its cross-architecture consistency, not on this ablation alone.

Cross-model: the lag replicates on an MoE architecture. We repeat the instrument validation and the band-wise steer on gpt-oss-20b (24 layers, MoE), with real decision points reconstructed from the same trajectories in a generic tool-choice format and bands mapped by depth-percentage (Table 3). The specificity control holds (few-shot two-hop baseline 14/20; answer flips 20/20 at motor vs random 1). The depth lag replicates: at the tail the verbalizable direction reroutes the answer (14/20 vs random 0) but not the action (0/40 vs random 0), and the action becomes specifically steerable only at the motor band (40/40 vs random 4; Fisher exact for the answer-tail and action-motor dissociations $p{=}3.3 \times 10^{-6}$ and $p{=}2.5 \times 10^{-18}$). The absolute onsets shift—on gpt-oss the answer consolidates later (tail, not mid-workspace) and the action later still (motor, not tail)—but the qualitative lag is the same across both architectures (Figure 1): a band where the answer is steerable and the action is not, with the action strictly deeper. gpt-oss's generic-format decision points have weaker edit-commitment fidelity (edit-points $P(\text{edit}){=}0.14$ vs bash-points 0.23); the random-controlled steer specificity is nonetheless clean.

[1] The dissociation column reports the edit→bash direction; the reverse bash→edit steer is weaker at the tail (44/60) and only completes at the motor band (60/60), consistent with the action requiring the deepest layers to fully commit.

model	band	answer →alt (rand)	action edit→bash (rand)
Qwen3.6-27B	mid	**11/20 (0)**	**0/60 (21)**
(dense, 64L)	tail	20/20 (0)	60/60 (23)
	motor	20/20 (0)	60/60 (21)
gpt-oss-20b	mid	1/20 (3)	0/40 (9)
(MoE, 24L)	tail	**14/20 (0)**	**0/40 (0)**
	motor	20/20 (1)	40/40 (4)

Table 3: The lag is cross-architecture. Bold marks each model's dissociation band—answer steerable, action not. The band's absolute depth is model-dependent (mid on Qwen, tail on gpt-oss); on both the action's steerability onset is strictly deeper than the answer's.

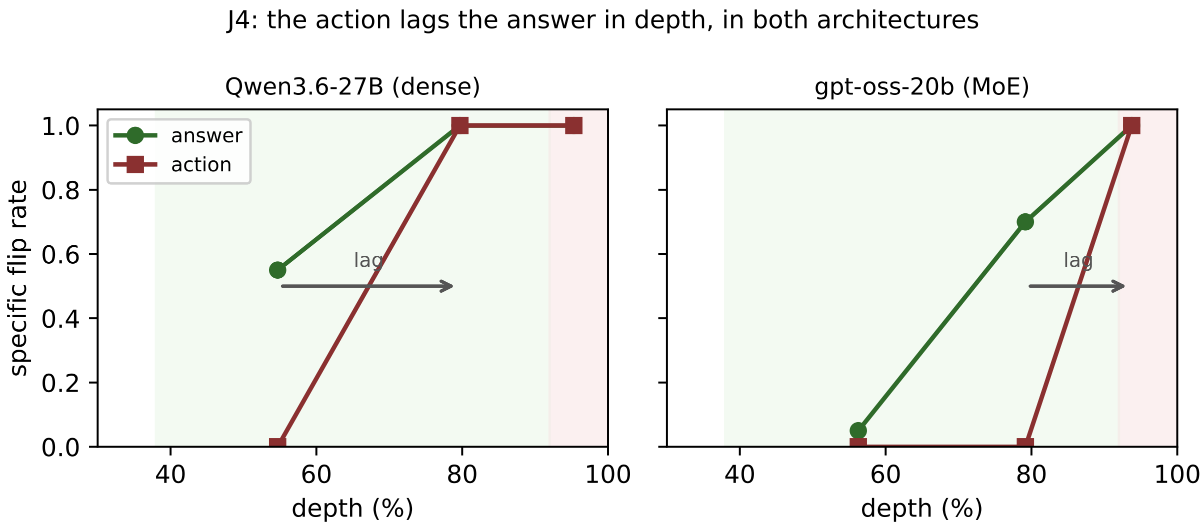

Figure 1: The reframe headline. In both architectures the verbalizable direction reroutes the *answer* at a shallower depth than it can reroute the *action* (green before red); the absolute band shifts with the model but the answer→action depth lag does not.

5 Discussion

Across a dense 27B and an MoE 20B, the verbalizable direction that Anthropic Interpretability Team [2026] steer to change an answer reaches the agent's action commitment *strictly later in depth* than it reaches the answer: there is a depth band where the held answer is causally steerable and the tool the agent commits to is not, and the commitment survives ablation of the verbalizable subspace at the decision point. This localizes the knowledge–action gap [Basu et al., 2026] as an answer→action *depth lag* and gives the arc's "lever is late" [Vicentino, 2026a,c] a reading in workspace terms.

Why the action lags (mechanism). Our prior circuit analysis of this agent [Vicentino, 2026b] identifies the commit as written by a small set of late attention heads (layer 59) that behave as induction/copy heads, promoting the tool-name token by copying it from the tool-call history rather than composing it from mid-network concepts. Such a late copy operation is inherently a "motor" computation, downstream of and orthogonal to the mid-network concept broadcast

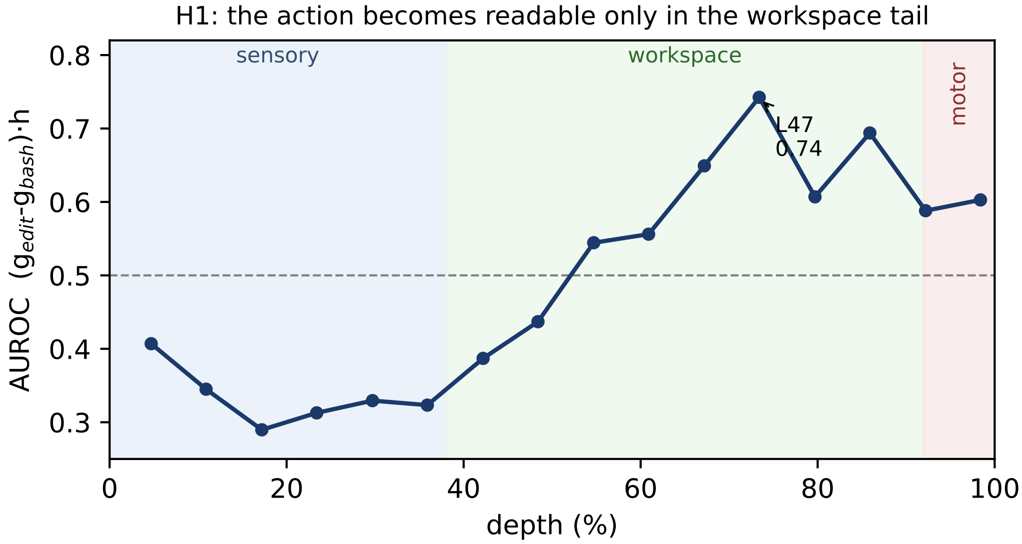

Figure 2: H1. The action's verbalizable readout (AUROC by depth) is near chance through most of mid-workspace and peaks at the tail (L47, 73%). Regimes after Anthropic Interpretability Team [2026].

that the verbalizable workspace steers—which is consistent with the lag we observe and with the commitment surviving ablation of the verbalizable subspace. A full head-level test is left to future work: prior analysis found layer-59 attention to be output-gated, so cleanly reconstructing per-head write directions there is nontrivial.

Monitoring implication (scoped). A J-lens-style verbalizable monitor reads a reasoning agent's answer/reasoning at a shallower depth—and in a band before—its action commitment: it would surface the model's stated intent while the committed act is still being decided deeper. This is scoped to verbalizable (J-lens) monitors on these two open-weights models; we make no architecture-general claim.

6 Limitations

Two open-weights models (a dense 27B and an MoE 20B), one behavior family (edit vs bash), n=60 action / 20 answer points on the primary model, prefill-only, single magnitude α=0.5 (controlled within each band by the random baseline). The depth lag replicates in both models but the *absolute* onset depths are model-dependent: on the dense model the answer becomes steerable at mid-network and the action only at the tail, whereas on the MoE model both shift deeper (answer at the tail, action at the motor band), so we frame the finding as a depth-relative lag that is consistent across the two architectures tested, not as a fixed "mid-workspace" band. On the MoE model the synthetic decision point is a weaker probe of the commit—baseline edit-commit fidelity 0.14 vs. bash 0.23—so its dissociation is corroborating rather than primary. The estimator is the row-restricted diagonal, not the full J-lens matrix, and we use a steer in place of the pre-registered coordinate swap (disclosed; the swap is a near-no-op in this basis; both are Anthropic-used and the steer is specificity-validated). Tool-related signals are linearly readable earlier than the commit depth in

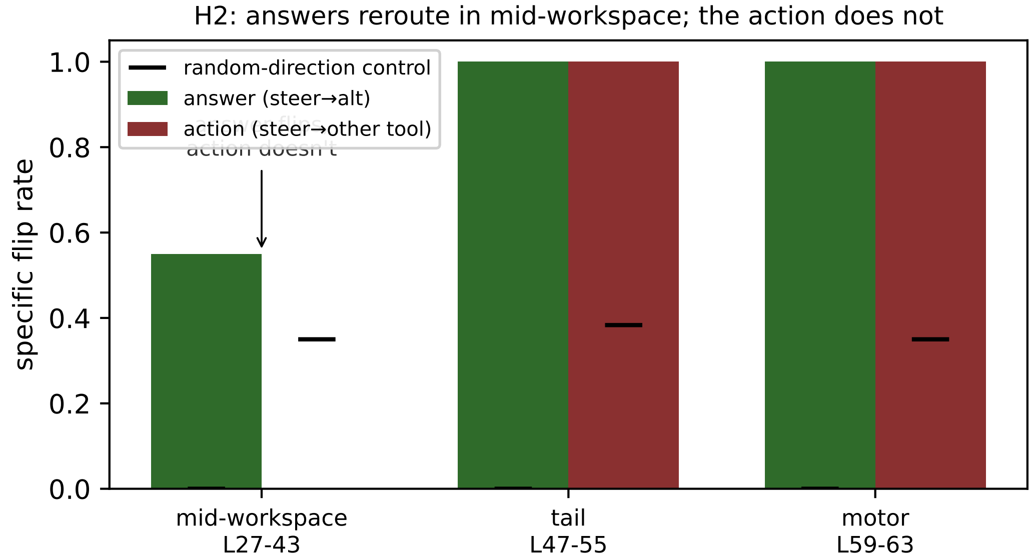

Figure 3: H2 dissociation. The verbalizable steer reroutes the *answer* in mid-workspace (11/20) but not the *action* (0/60, below the random control); both converge at the tail/motor. Black ticks: random-direction controls.

general [Sun & Kazakov, 2026, Sun et al., 2026]; our claim is about specific causal steerability of the commitment via the verbalizable direction, not readability. No claim of architecture-generality beyond these two models; the workspace paper's own agent generality is untested.

7 Reproducibility

Public HF ledgers (`results/jspace_action.json` for the dense model, `results/jspace_xmodel_gpt-oss-20b.json` for the MoE), estimated vectors, and decision-point residuals; harnesses `scripts/jspace_action.py`, `scripts/jspace_xmodel.py` and pre-registration released. The evaluation `scripts/eval_jspace.py` recomputes the H1 readout offline from the public vectors and residuals (16/16 layers to 10^{-6}) and audits the dissociation logic for both models. Steer and ablation numbers require GPU generation; an independent reimplementation of the steer/ablation pipeline, run with fresh random seeds, reproduced all six targeted dissociation counts exactly (answers 11/20/20, actions 0/60/60 at mid/tail/motor) and the ablated differential (0.167)—the targeted counts are seed-independent while only the random controls vary—and both are logged to the public ledger (`verify` field).

References

Anthropic Interpretability Team. Verbalizable Representations Form a Global Workspace in Language Models. *Transformer Circuits Thread*, 2026. transformer-circuits.pub/2026/workspace.

S. Basu et al. Interpretability Without Actionability: Mechanistic Methods Cannot Correct Language Model Errors Despite Near-Perfect Internal Representations. *arXiv:2603.18353*, 2026.

C.-E. Sun, L. Liu, G. Yan, Z. Wang, T.-W. Weng. LLM Agents Already Know When to Call Tools — Even Without Reasoning. *arXiv:2605.09252*, 2026.

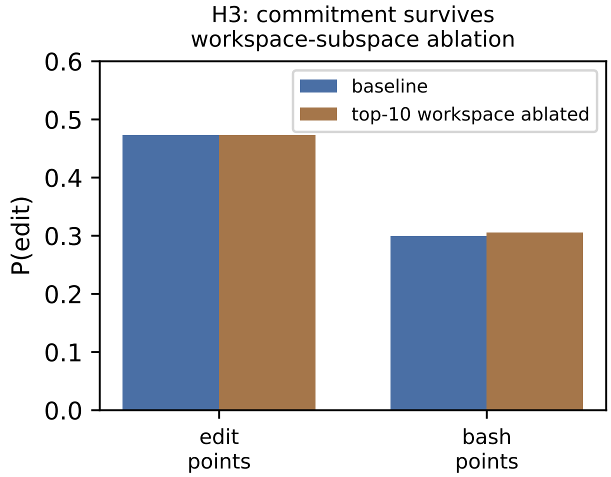

Figure 4: H3. Ablating the top-10 verbalizable directions at the decision point leaves the commitment differential intact.

T. Sun and D. Kazakov. Tool-Call Dependency Structure is Linearly Decodable in LLM Agent Residual Streams. *arXiv:2605.25310*, 2026.

Z. Wu, Z. Wang, S. Cho, Y. Yang, A. Koshiyama, S. Bulathwela, M. Perez-Ortiz. Tool Calling is Linearly Readable and Steerable in Language Models. *arXiv:2605.07990*, 2026.

Z. Cui and X. Luo. Do Agents Think Deeper? A Mechanistic Investigation of Layer-Wise Dynamics in Sequential Planning. *arXiv:2605.27935*, 2026.

A. Goyal et al. Coordination Among Neural Modules Through a Shared Global Workspace. *arXiv:2103.01197*, 2021.

R. VanRullen and R. Kanai. Deep Learning and the Global Workspace Theory. *Trends in Neurosciences*, 2021. arXiv:2012.10390.

C. Vicentino. The Lever Is Late: the Knowledge–Action Gap Is a Layer Gap. *Zenodo*, 2026. DOI 10.5281/zenodo.20534219.

C. Vicentino. The Commitment Lever Generalizes — and It Brakes. *Zenodo*, 2026. DOI 10.5281/zenodo.20679287.

C. Vicentino. The Late Channel: Chain-of-Thought Is Causal Only When It Changes the Outcome. *Zenodo*, 2026. DOI 10.5281/zenodo.20752895.